

Ngo Hoai Nam

+84 901 620 311 | hmansocieties11@gmail.com

[ngo-hoai-nam](https://www.linkedin.com/company/ngo-hoai-nam) | [justHman](https://www.github.com/justHman)

Ho Chi Minh City, Vietnam

OBJECTIVE

Passionate AI Major at FPT University seeking challenging opportunities to leverage expertise in computer vision, natural language processing, sound and machine learning. Aiming to contribute to innovative projects that build transparent, data-driven systems making tangible social impact through artificial intelligence solutions.

EXPERIENCE

- Freelance Mozilla Addon Developer** 2025
AI-Powered Browser Extension Developer Remote
 - Developed automated Mozilla Firefox addon creation system using AI agents and LLM models
 - Implemented end-to-end automation pipeline: topic generation, code generation, and logo creation
 - Utilized Large Language Models for automatic code generation and browser extension functionality
 - Integrated AI image generation models for automatic logo and visual asset creation
 - Created automated packaging system to zip completed addons for Mozilla marketplace distribution
 - Delivered multiple functional browser extensions with minimal manual intervention
- 2025 NASA Space Apps Challenge** [🌟] 2025
Team Leader & Solution Architect VNG, District 7, Ho Chi Minh City
 - Led team in "A World Away - Hunting for Exoplanets with AI" challenge
 - Developed AI solutions for exoplanet detection using NASA data with 3 classification labels
 - Provided architectural models and structured data solutions for galactic problem solving
 - Applied machine learning techniques to analyze astronomical datasets
- LEGAL HACKATHON 2025** [🌟] 2025
Lead Developer & Web Engineer UEH, Ho Chi Minh City
 - Developed web application for video-to-audio conversion and sound pattern visualization
 - Implemented solution to detect secret filming in cinemas using audio analysis
 - Utilized Streamlit, HTML/CSS, Librosa, and Matplotlib for comprehensive audio processing
 - Achieved Third Prize for innovative approach to cinema security challenges
- FPT HACKATHON 2025: CHIP AI - LOGISTICS ON EDGE** [🌟] 2025
Contestant FPTU, District 9, Ho Chi Minh City
 - Participated in ICPC format competition programming challenges
 - Developed computer vision solutions for logistics shipping optimization problems
 - Applied AI algorithms to enhance edge computing for logistics operations
 - Collaborated with team to solve complex supply chain management scenarios
- AI INNOVATION HACKATHON SUMMER 2025** [🌟] 2025
Contestant FPTU, District 9, Ho Chi Minh City
 - Built comprehensive AI applications for course search and academic project management
 - Implemented AI stack including Face Recognition, Recommendation System, and Chatbot
 - Developed solutions for previous semester project optimization and discovery
 - Demonstrated proficiency in multiple AI technologies integration
- F TALENT CODE 2024** [🌟] 2024
Contestant Student Cultural House, Dong Hoa, Di An, Ho Chi Minh City
 - Competed in ICPC standard format programming competition
 - Demonstrated algorithmic problem-solving and competitive programming skills
 - Applied data structures and algorithms knowledge under time constraints
 - Enhanced coding proficiency through structured programming challenges

EDUCATION

- FPT University** 2023 - Present
Bachelor of Science in Artificial Intelligence Ho Chi Minh City, Vietnam
 - Major: AI with focus on Computer Vision, NLP, Sound and Machine Learning
 - GPA: 8.56/10

PROJECTS

- **PPE Detection System**
Tools: YOLOv8, Python, OpenCV, Computer Vision
 - Developed YOLOv8-based system for detecting protective gear (helmets, vests, gloves,...) on workers
 - Implemented safety-check algorithm to determine worker safety status in real-time
 - Created interactive user interface for real-time monitoring and visualization of detection outcomes
 - Applied computer vision techniques for workplace safety compliance monitoring
 - **Real-time Face Recognition & Attendance System**
Tools: OpenCV, PCA, Python, Computer Vision
 - Developed real-time face recognition system using Principal Component Analysis (PCA)
 - Implemented automatic attendance tracking with high accuracy and real-time processing
 - Created face data collection and model training pipeline for recognition system
 - Developed user-friendly interface for attendance management and logging
 - **Natural Food Vertical Search Engine**
Tools: TF-IDF, FAISS, Python, Information Retrieval
 - Built inverted index using TF-IDF and FAISS for semantic search on Natural Food corpus
 - Achieved high-speed search performance (<1s) with semantic understanding capabilities
 - Integrated spelling correction and user-friendly interface with result snippets
 - Implemented multiple search methods for enhanced user experience
 - **Vietnamese Image Captioning System**
Tools: BLIP, PhoBERT, PyTorch, Transformers
 - Fine-tuned image captioning models on Vietnamese datasets (KTVIC and UITVIC)
 - Implemented both accented and unaccented approaches for Vietnamese language processing
 - Used PhoBERT tokenization to enhance Vietnamese language understanding
 - Developed transfer-learning and train-from-scratch methodologies for caption generation

2024



2024



2024



2024


- SKILLS
- **Programming Languages:** Python, C++, C#, HTML5, CSS3
 - **Web Frameworks & UI:** Streamlit, Flask, Django, FastAPI, Tkinter
 - **Database Systems:** MySQL, MongoDB
 - **AI & Machine Learning:** PyTorch, TensorFlow, OpenCV, Transformers, scikit-learn, FAISS
 - **Data Analysis & Visualization:** NumPy, pandas, Matplotlib, Seaborn, Plotly, Folium
 - **Development Tools:** Git, VS Code, PyCharm, Thonny
 - **Computer Vision & NLP:** Image Captioning, Object Detection, Face Recognition, Text Processing
 - **Mathematical & Statistical Tools:** NumPy, SciPy, Statsmodels, MATLAB, R
 - **Other Tools & Technologies:** Selenium, BeautifulSoup, Librosa, autoclick, 2FA
 - **Research Areas:** Computer Vision, Natural Language Processing, Sound, Machine Learning, Deep Learning, Information Retrieval, AI Systems
- HONORS AND AWARDS
- **LEGAL HACKATHON 2025 - Third Prize**
University of Economics Ho Chi Minh City (UEH)
 - Built web application for video upload, audio extraction, and sound pattern visualization
 - Developed innovative solution for detecting secret filming in cinemas
 - Demonstrated expertise in audio processing and web development technologies
 - **Traditional Musical Instrument Competition**
FPT University
 - Participated in both solo and band categories for traditional Vietnamese instruments
 - Promoted national culture through musical performances
 - Demonstrated cultural awareness and performance skills

2025



2024 - 2025



VOLUNTEER EXPERIENCE

• Traditional Musical Instrument Performer FPT University ◦ Perform traditional Vietnamese musical instruments in high schools and public venues ◦ Promote national culture and heritage to young people and international audiences ◦ Contribute to cultural preservation and education through artistic expression	2023 - Present 
• English Teaching Volunteer English Language FPTU (ELF) ◦ Provided free English language instruction to children both online and offline ◦ Developed systematic curriculum for English language learning ◦ Enhanced communication skills and educational methodology experience	2024 

CERTIFICATIONS

• IBM AI Developer	2024
• IBM: AI Foundations for Everyone	2024
• University of Sydney: Academic Skills for University Success	2024
• IBM: Generative AI and LLMs: Architecture and Data Preparation	2025
• IBM Data Science Fundamentals with Python and SQL	2025
• IBM Full Stack Software Developer	2025
• LearnQuest: Ethics in the Age of AI	2025
• University of Minnesota: Software Development Lifecycle	2025
• KS57 1 Digital Ecosystem: From Governance to Business	2025
• KS57 2 Key Activities of Digital Transformation	2025

ADDITIONAL INFORMATION

Languages: Vietnamese (Native), English (Fluent), Japanese (Basic)
Interests: Artificial Intelligence Research, Computer Vision, Natural Language Processing, Open Source Development, Automation Systems, Robotics, AIoT, Sound, Music

REFERENCES

1. **Mr. Đỗ Đức Hòa**

CF Department FU-HCM

FPT University, Ho Chi Minh City, Vietnam

Email: haodd3@fe.edu.vn

Phone: +84 909 441 246

Relationship: Academic Advisor
2. **Mr. Hồ Lê Minh Toàn**

Lecturer at FPT University

FPT University, Ho Chi Minh City, Vietnam

Email: holeminhtoan@gmail.com

Relationship: Academic Advisor
3. **Mr. Vũ Hà Anh**

Lecturer at FPT University

FPT University, Ho Chi Minh City, Vietnam

Email: anhv54@fe.edu.vn

Phone: +84 366 765 835

Relationship: Academic Advisor